Rising Sun: Light Test

Steps for David

Dr. O decided everything on this sheet. Do each step in order. Do not skip a step. Do not change anything. If something does not match what the step says, STOP and call Dr. O. Do not guess.

When you finish, take one photo and send it to Dr. O. That is the whole report.

Before you start: check you have these

[ ] The ESP32-S3 board (the same kind as the Holopod buttons).

[ ] The LED strip, 1 meter.

[ ] The 5 volt power supply for the strip.

[ ] One 330 ohm resistor.

[ ] One 1000 microfarad capacitor (if Dr. O gave you one).

[ ] Jumper wires.

[ ] The USB cable that came for the board.

[ ] A computer with the Arduino program.

[ ] The file sunrise.ino from Dr. O.

STOP. If anything above is missing, call Dr. O. Do not start.

Part 1: Put the file ready

Step 1. Open the Arduino program.

Step 2. Open the file sunrise.ino.

Step 3. Find the line that says: const int LED_COUNT = 60;

Step 4. Dr. O writes the number of LEDs on your strip here: _____ . Type that number in place of 60. Change nothing else on that line.

Step 5. Find the line that says: const bool DEMO_MODE = true; It must say true. Do not change it.

Part 2: Wire it. The power is OFF for all of this.

Unplug the power supply and the USB cable before you start.

Step 6. Look at the strip. One end has three connection points: 5V, DIN (also written DI), and GND. Use that end. The little arrows on the strip point away from it.

Step 7. Take the 330 ohm resistor. Connect one leg to the strip wire marked DIN.

Step 8. Connect the other leg of the resistor to the pin on the board that says 5 (it may say GPIO5).

Step 9. Connect the strip wire marked GND to a GND pin on the board.

Step 10. Connect the strip wire marked GND again to the minus side of the power supply.

Step 11. Connect the strip wire marked 5V to the plus side of the power supply. Do NOT connect it to the board.

STOP. If you cannot find a pin marked 5, or the strip has different markings than 5V, DIN and GND, call Dr. O.

Part 3: Load the program

Step 12. Plug the USB cable into the board and into the computer.

Step 13. In the Arduino program, press the Upload button (the arrow).

Step 14. Wait until it says Done uploading.

STOP. If it shows a red error message, take a photo of the message and send it to Dr. O. Do not try to fix it.

Part 4: Turn it on

Step 15. Plug in the power supply for the strip.

Step 16. Watch the strip for two minutes.

What you should see

A quick glow up and down. Then the strip slowly goes from dark, to dim red, to orange, to yellow, to bright warm white. It takes about 90 seconds. Then it goes dark for 10 seconds and starts again.

STOP. If the strip does not light at all, or the first light flickers, or the colors are very different from the list above, take a photo and a short video and send them to Dr. O. Do not change anything.

Part 5: Send the photo

Step 17. Hold the strip lit, bright warm white, and take one photo.

Step 18. Send the photo to Dr. O.

Step 19. Unplug the power supply first, then the USB cable.

You are done. Do not open any other file. Do not change any other setting.